

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.			
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.			
20							***** MODULAR VALVES *****							
30										817	USE ONLY FOR TIE IN			
30										B1	ALL BOLTS ARE METRIC DIA.			
40	U04		MODULAR DBB	K25RT02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
40			MODULAR DBB	K25RT08	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV	0	0					
50		VAV, 817	MODULAR DBB	K25CT08	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV	0	0					
60	U04	VAW	MODULAR DBB	M25RT02	0.75	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
70			MODULAR DBB	M25RT08	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES			
80	U04	VAW	MODULAR DBB	M25HT02	0.75	1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	UI2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
90	U04	VAW	MODULAR DBB	M25HT02	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
100	U04	VAU, 817	MODULAR DBB	M25YT02	1	1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE R2BVR	3	3	VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END			
110	U04	VAV, 817	MODULAR DBB	K25NT08	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF X RTJ SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	VAU				
120	U04	VAV, 817	MODULAR DBB	K25NT08L	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF X RTJ SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	VAU	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
130	U04	VAU, 817	MODULAR DBB	M25YT02L	1	1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE R2BVR	3	3	VAU				
140	U04	VAW	MODULAR DBB	M25HT02L	0.75	1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
150	U04	VAW	MODULAR DBB	M25HT02L	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
160		VB2					***** BALL VALVES *****							
170														
170			BALL	A25RT07	2	2	BALL 2500# RTJ SS A182 F321 316SS TRIM TRUNNION SOFT RB GO	0	0					
180	U04	817	BALL	A25CT07	3	4	BALL 2500# CF SS A182 F321 316SS TRIM TRUNNION SOFT RB GO	0	0					
190			BALL	B25RT01	0.75	1.5	BALL 2500# RTJ SS A182 F321 316SS TRIM FLOATING SOFT FB LO	0	0					
200		VA5	BALL	B25RT07	2	2	BALL 2500# RTJ SS A182 F321 316SS TRIM TRUNNION SOFT FB GO	0	0					
210	U04	817	BALL	B25CT07	3	3	BALL 2500# CF SS A182 F321 316SS TRIM TRUNNION SOFT FB GO	0	0					
220														
220							***** GLOBE VALVES *****							
230														
230			GLOBE	L25RT48	0.75	1.5	GLOBE 2500# RTJ SS A182 F321 TRIM 16 OS&Y HW	0	0					
240														
240							***** CHECK VALVES *****							
250														
250		VC1	CHECK	C25RT50	0.75	1.5	CHECK PISTON 2500# RTJ SS A182 F321 TRIM 16	0	0					
260	U04		CHECK	C25RT52	2	2	CHECK DUAL 2500# RTJ SS A182 F321 TRIM 16	0	0					
270	U04	817	CHECK	C25CT52	3	3	CHECK DUAL DBL FLG 2500# CF SS A182 F321 TRIM 16	0	0					
280														
280							***** PIPE *****							
290														
290			PIPE	5365202	0.75	1.5	PIPE SCH XXS SMLS SS A312 316/316L SS PE	0	0					
300			PIPE	5365208	2	2	PIPE WT 12.7MM SMLS SS A312 316/316L BE	0	0					
310			PIPE	5365209	3	3	PIPE WT 19.05MM SMLS SS A312 316/316L BE	0	0					
320			PIPE	5365210	4	4	PIPE WT 22.00MM SMLS SS A312 316/316L BE	0	0					
330			PIPE	5365211	6	6	PIPE WT 24.00MM SMLS SS A312 316/316L BE	0	0					
340		812	PIPE	5365206	0.75	0.75	PIPE SCH XXS SMLS SS A312 316/316L SS BE	0	0					
350														
350							***** FITTINGS *****							
360														
360			ELBOW	5449141	0.75	1.5	ELL BW 90 DEG LR SCH XXS SS A403 WP316/316L-S	0	0					
370			ELBOW	5449057	2	2	ELL BW 90 DEG LR WT 12.7MM SS A403 WP316/316L-S	0	0					
380			ELBOW	5449058	3	3	ELL BW 90 DEG LR WT 19.05MM SS A403 WP316/316L-S	0	0					
390			ELBOW	5449082	4	4	ELL BW 90 DEG LR WT 22.00MM SS A403 WP316/316L-S	0	0					
400			ELBOW	5449143	0.75	1.5	ELL 45 DEG LR XXS SS A403 WP316/316L-S	0	0					
410			ELBOW	5449059	2	2	ELL BW 45 DEG LR WT 12.7MM SS A403 WP316/316L-S	0	0					
420			ELBOW	5449060	3	3	ELL BW 45 DEG LR WT 19.05MM SS A403 WP316/316L-S	0	0					
430			ELBOW	5449083	4	4	ELL BW 45 DEG LR WT 22.00MM SS A403 WP316/316L-S	0	0					
440														
440			TEE	5449191	0.75	1.5	TEE SCH XXS SS A403 WP316/316L-S	0	0					
450			TEE	5449061	2	2	TEE WT 12.7MM SS A403 WP316/316L-S	0	0					
460			TEE	5449062	3	3	TEE WT 19.05MM SS A403 WP316/316L-S	0	0					
470			TEE	5449183	4	4	TEE WT 22.00MM SS A403 WP316/316L-S	0	0					
480			TEE RED	5449192	0.75	1	TEE RED SCH XXS SS A403 WP316/316L-S	1	1.5					
			TENGIZCHEVROIL TCO, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
					ALL SIZES - BUTT WELDED SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.4 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS (F321 FLANGE) DUAL GRADE FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -135 / 120°C	12/16/2024	BY OZAC	U04	Page 1 of 3	2500P2

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
490			TEE RED	5449063	0.75	1.5	TEE RED WT 12.7MM X SCH XXS SS A403 WP316/316L-S	2	2	
500			TEE RED	5449064	1.5	1.5	TEE RED WT 19.05MM X SCH XXS SS A403 WP316/316L-S	3	3	
510			TEE RED	5449065	2	2	TEE RED WT 19.05MM/WT 12.7 MM SS A403 WP316/316L-S	3	3	
520			TEE RED	5449029	2	2	TEE RED WT 22.0MM/WT 12.7 MM SS A403 WP316/316L-S	4	4	
530			TEE RED	5449098	3	3	TEE RED WT 22.0MM/WT 19.05 MM SS A403 WP316/316L-S	4	4	
540										
540			CAP	5449056	0.75	1.5	CAP SCH XXS SS A403 WP316/316L-S	0	0	
550			CAP	5449066	2	2	CAP WT 12.7MM SS A403 WP316/316L-S	0	0	
560			CAP	5449067	3	3	CAP WT 19.05MM SS A403 WP316/316L-S	0	0	
570			CAP	5449099	4	4	CAP WT 22.00MM SS A403 WP316/316L-S	0	0	
580			CAP	5449100	6	6	CAP WT 24.00MM SS A403 WP316/316L-S	0	0	
590										
590			RED CONC	5449181	0.75	1	REDUCER CONC SCH XXS SS A403 WP316/316L-S	1	1.5	
600			RED CONC	5449068	0.75	1.5	REDUCER CONC WT 12.7MM/SCH XXS SS A403 WP316/316L-S	2	2	
610			RED CONC	5449069	1.5	1.5	REDUCER CONC WT 19.05MM/SCH XXS SS A403 WP316/316L-S	3	3	
620			RED CONC	5392365	1.5	1.5	REDUCER CONC WT 22.0MM/ SCH XXS SS A403 WP316/316L-S	4	4	
630			RED CONC	5449070	2	2	REDUCER CONC WT 19.05MM/WT 12.7MM SS A403WP316/316L-S	3	3	
640			RED CONC	5392349	2	2	REDUCER CONC WT 22.0MM/ 12.7MM SS A403 WP316/316L-S	4	4	
650			RED CONC	5449074	3	3	REDUCER CONC WT 22.0MM/19.05MM SS A403 WP316/316L-S	4	4	
660			RED ECC	5449186	0.75	1	REDUCER ECC SCH XXS SS A403 WP316/316L-S	1	1.5	
670			RED ECC	5449071	0.75	1.5	REDUCER ECC WT 12.7MM/SCH XXS SS A403 WP316/316L-S	2	2	
680			RED ECC	5449072	1.5	1.5	REDUCER ECC WT 19.05MM / SCH XXS SS A403 WP316/316L-S	3	3	
690			RED ECC	5449182	1.5	1.5	REDUCER ECC WT 22.0MM / SCH XXS SS A403 WP316/316L-S	4	4	
700			RED ECC	5449073	2	2	REDUCER ECC WT 19.05MM / WT 12.7MM SS A403WP316/316L-S	3	3	
710			RED ECC	5449188	2	2	REDUCER ECC WT 22.0MM / 12.7MM SS A403 WP316/316L-S	4	4	
720			RED ECC	5449075	3	3	REDUCER ECC WT 22.0MM / 19.05MM SS A403 WP316/316L-S	4	4	
730							***** OLETS *****			
740										
740		UJ2	OLETS	5504586	0.75	1.5	NIPOFLANGE 2500# RTJ WT 12.7MM / SCH XXS SS A182 F321	2	2	
750		UJ2	OLETS	5504587	0.75	1.5	NIPOFLANGE 2500# RTJ WT 19.05 MM / SCH XXS SS A182 F321	3	3	
760		UJ2	OLETS	5504588	0.75	1.5	NIPOFLANGE 2500# RTJ WT 22.00MM/XXS SS A182 F321	4	4	
770	U04		OLETS	5504589	0.75	1	NIPOFLANGE 2500# RTJ WT 19.05 MM / SCH XXS SS A182 F321	3	3	
780		FF5	OLETS	5504610	0.75	0.75	NIPOFLANGE FLAT BOTTOM 2500# RTJ SCH XXS SS A182 F321	0	0	
790										
790			OLETS	5509662	0.75	1	WELDOLET19.05MM WT/SCH XXS SS A182 F316/316L	3	3	
800										
800			OLETS	5509663	0.75	1.5	WELDOLET 22.00MM WT/SCH XXS SS A182 F316/316L	4	4	
810			OLETS	5509664	2	2	WELDOLET 24.00MM WT/ 12.7MM SS A182 F316/316L	6	6	
820	U04		OLETS	5504378	0.75	0.75	WELDOLET FLAT BOTTOM SCH XXS SS A182 F316/F316L	0	0	
830										
830		UJ2	ELBOFLG	5561561	1.5	1.5	ELBOFLANGE 2500# RTJ WT 12.7MM / SCH XXS SS A182 F321	2	2	
840		UJ2	ELBOFLG	5561562	1.5	1.5	ELBOFLANGE 2500# RTJ WT 19.05MM / SCH XXS SS A182 F321	3	3	
850		UJ2	ELBOFLG	5561563	1.5	1.5	ELBOFLANGE 2500# RTJ WT 22.0MM / SCH XXS SS A182 F321	4	4	
860							***** FLANGES *****			
870		816								
870			FLANGE	5584161	0.5	1.5	FLG WN 2500# RTJ SS A182 F321 SCH XXS	0	0	
880			FLANGE	5584162	2	2	FLG WN 2500# RTJ SS A182 F321 WT 12.7MM	0	0	
890			FLANGE	5584163	3	3	FLG WN 2500# RTJ SS A182 F321 WT 19.05MM	0	0	
900	U04	817	FLANGE	5619867	3	3	FLG COMPACT 2500# RJ SS A182 F321 WT 19.05MM	0	0	
910			FLANGE	5584274	3	3	FLG WN 2500# RF SS A182 F321 WT 19.05MM	0	0	
920			FLANGE	5584275	4	4	FLG WN 2500# RF SS A182 F321 WT 22.0MM	0	0	
930		FA8	FLANGE	5599831	0.5	0.5	FLG SCR D (FNPT) REDUCING 2500# RTJ SS A182 F321	0.75	0.75	
940										
940			FLANGE	5851213	0.75	0.75	FLANGE ADAPTER 2500# RTJ 321 SS X 12 MM TUBE	0	0	
950			FLANGE	5851213	2	2	FLANGE ADAPTER 2500# RTJ 321 SS X 12 MM TUBE	0	0	
960										
960			BLIND	5616584	3	4	FLG BLIND 2500# RF SS A182 F321	0	0	
970			BLIND	5619738	0.75	2	FLG BLIND 2500# RTJ SS A182 F321	0	0	
980	U04	817	FLANGE	5619868	4	4	FLG COMPACT 2500# RJ SS A182 F321 WT 22.0MM LINE CLASS 2500P2	0	0	
990	U04	817	BLIND	5619798	3	3	FLG COMPACT BLIND 2500# RJ SS A182 F321 PIPE CLASS 2500P2	0	0	
1000							***** ORIFICE FLANGES *****			
1000										
1010										
1010	U04		ORIFICE FLG	5642051	3	3	WN ORIF SET 2500# RTJ SS A182 F321 WT 19.05MM	0	0	
1020							***** GASKETS *****			
1020										
1030										
1030			GASKET	5672272	0.5	4	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR MANUFACTURERS STANDARD	0	0	
1040			GASKET	5674480	0.5	3	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
1050	U04	817	GASKET	5673531	3	4	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500P2	0	0	
1060										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELDED SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.4 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS (F321 FLANGE) DUAL GRADE FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -135 / 120°C	12/16/2024	BY OZAC	U04	Page 2 of 3	2500P2

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1060		B1					***** BOLTING *****			
1070										
1070	U04		BOLTING	5675770	0.5	4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1080										
1080							***** BLINDS & SPACERS *****			
1090										
1090			SPEC BLIND	5749592	1	2	SPECTACLE BLIND 2500# RTJ SS A240 GR 316	0	0	
1100	U04		SPEC BLIND	5749592	3	3	SPECTACLE BLIND 2500# RTJ SS A240 GR 316	0	0	
1110	U04	817	SPEC BLIND	5749573	3	4	SPECTACLE BLIND COMPACT 2500# RJ SS A240 GR 316	0	0	
1120										
1120	U04	817	PADDLE	5407808	3	4	PADDLE BLANK COMPACT 2500# RJ SS A240 GR 316	0	0	
1130										
1130			SPEC BLIND	5749514	3	4	SPECTACLE BLIND 2500# RF SS A240 GR 316	0	0	
1140										
1140	U04	817	PADDLE	5407807	3	4	PADDLE SPACER COMPACT 2500# RJ SS A240 GR 316	0	0	
1150										

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